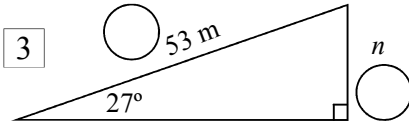
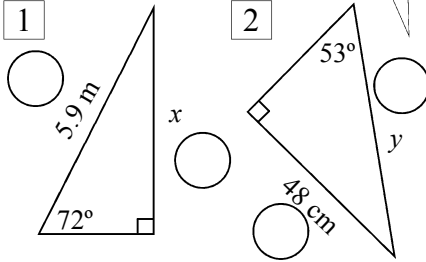


Trigonometry - Finding a Side (sin) (Sheet 1)

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Label these sides 'O', and 'H'.



Calculate these, answer to 1 d.p.

4 $\frac{13}{\sin 21^\circ} = \underline{\hspace{2cm}}$ 5 $\frac{54.7}{\sin 77^\circ} = \underline{\hspace{2cm}}$

Calc: $\boxed{1}\boxed{3}\boxed{\div}\boxed{\sin}\boxed{2}\boxed{1}\boxed{=}$

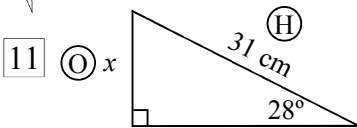
6 $94 \sin 36^\circ = \underline{\hspace{2cm}}$
Calc: $\boxed{9}\boxed{4}\boxed{\sin}\boxed{3}\boxed{6}\boxed{=}$

7 $11.2 \sin 62^\circ = \underline{\hspace{2cm}}$

8 $118 \sin 48^\circ = \underline{\hspace{2cm}}$

9 $\frac{200}{\sin 55^\circ} = \underline{\hspace{2cm}}$ 10 $\frac{0.8}{\sin 32^\circ} = \underline{\hspace{2cm}}$

Calculate the side length.
Answering to 1 decimal place.

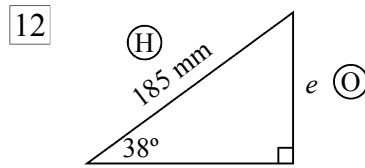


$$\sin 28 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 31 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 31$$

$$x = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$x = \boxed{\hspace{1cm}} \text{ cm } 1 \text{ d.p.}$$

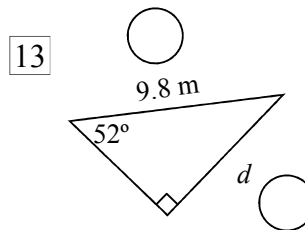


$$\sin 38 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 185 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 185$$

$$e = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$e = \boxed{\hspace{1cm}} \text{ mm } 1 \text{ d.p.}$$



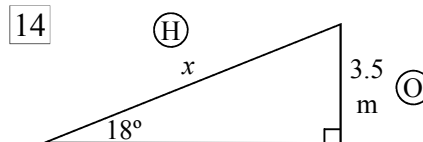
$$\sin 52 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 9.8 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 9.8$$

$$d = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$d = \boxed{\hspace{1cm}} \text{ m } 1 \text{ d.p.}$$

Now 'x' (the term) is on the bottom

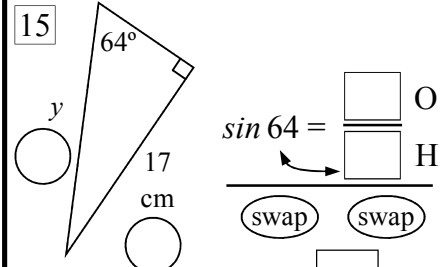


$$\sin 18 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 3.5 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 3.5$$

$$x = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$x = \boxed{\hspace{1cm}} \text{ m } 1 \text{ d.p.}$$

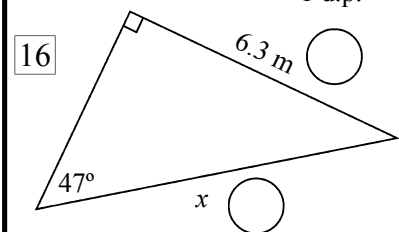


$$\sin 64 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 17 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 17$$

$$y = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$y = \boxed{\hspace{1cm}} \text{ cm } 1 \text{ d.p.}$$

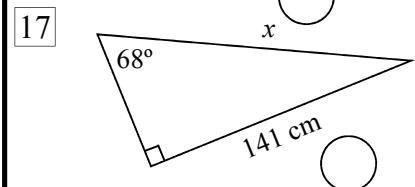


$$\sin 47 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 6.3 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 6.3$$

$$x = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$x = \boxed{\hspace{1cm}} \text{ m } 1 \text{ d.p.}$$



$$\sin 68 = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \text{ O } \text{ H }$$

$$\frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 141 \quad \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}} \times 141$$

$$x = \boxed{\hspace{1cm}} \sin \boxed{\hspace{1cm}}$$

$$x = \boxed{\hspace{1cm}} \text{ cm } 1 \text{ d.p.}$$

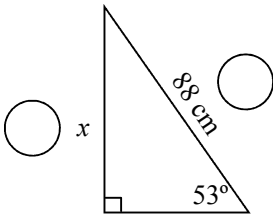
Trigonometry - Finding a Side (sin) (Sheet 2)

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Find the side length (1 d.p.).
All are sin, but are a mix of 'x on top' and 'x on the bottom'.

1



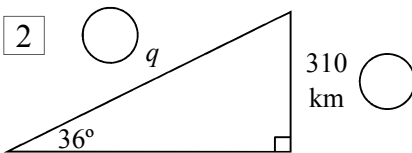
$$\sin \frac{\text{O}}{\text{H}} = \frac{\text{O}}{\text{H}}$$

(x 88) (x 88)

$$x = \sin$$

$$x = \text{cm} \quad 1 \text{ d.p.}$$

2



$$\sin \frac{\text{O}}{\text{H}} = \frac{\text{O}}{\text{H}}$$

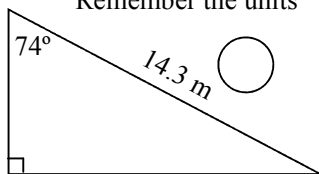
(swap) (swap)

$$q = \frac{\sin}{\text{H}}$$

$$q =$$

Remember the units

3

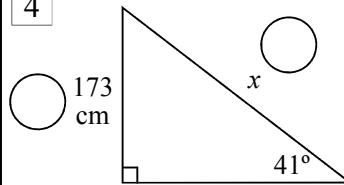


$$\sin \frac{\text{O}}{\text{H}} = \frac{\text{O}}{\text{H}}$$

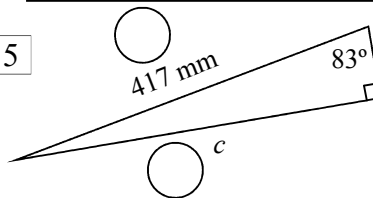
$$k = \sin$$

$$k =$$

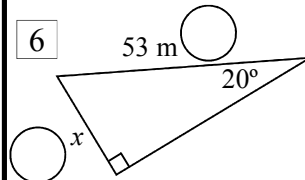
4



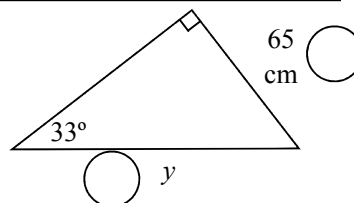
5



6



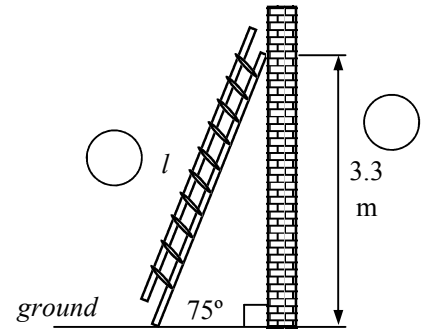
7



Answer these using your sin trig skills. Find the lengths to 1 decimal place.



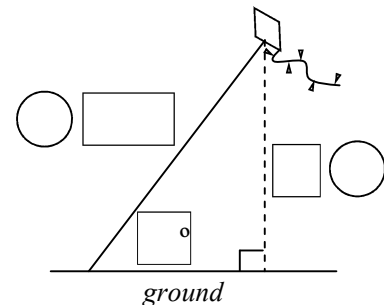
A ladder is leaning against a wall making an angle of 75° with the ground. The ladder reaches 3.3m up the wall.



8 Find 'l', the length of the ladder.

A kite string is 55 m long and is fully extended. The string makes an angle of 64° with the ground.

9 Write 'x' as the height of the kite. Then write the string length and angle in the two boxes on the diagram



10 Label the distances 'O' and 'H' then calculate the height of the kite.